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Advanced Math

10

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

14

The correct answer is 14. It's given by the first equation of the system of equations that $y = -2.5$. Substituting -2.5 for y in the second given equation, $y = x^2 + 8x + k$, yields $-2.5 = x^2 + 8x + k$. Adding 2.5 to both sides of this equation yields $0 = x^2 + 8x + k + 2.5$. A quadratic equation of the form $0 = ax^2 + bx + c$, where a, b, and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. In the equation $0 = x^2 + 8x + k + 2.5$, where k is a positive integer constant, $a = 1$, $b = 8$, and $c = k + 2.5$. Substituting 1 for a, 8 for b, and $k + 2.5$ for c in $b^2 - 4ac$ yields $8^2 - 41k + 2.5$, or $64 - 4k + 2.5$. Since this value must be negative, $64 - 4k + 2.5 < 0$. Adding $4k + 2.5$ to both sides of this inequality yields $64 < 4k + 2.5$. Dividing both sides of this inequality by 4 yields $16 < k + 2.5$. Subtracting 2.5 from both sides of this inequality yields $13.5 < k$. Since k is a positive integer constant, the least possible value of k is 14.

Q2

D

D. 8

Choice D is correct. It's given that the graph of $y = h(x)$ intersects the x-axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x-axis when $y = 0$ or when $h(x) = 0$, it follows that $h(0) = 0$ and $h(t) = 0$. If $h(t) = 0$, then $0 = 2(t-4)^2 - 32$. Adding 32 to both sides of this equation yields $32 = 2(t-4)^2$. Dividing both sides of this equation by 2 yields $16 = (t-4)^2$. Taking the square root of both sides of this equation yields $4 = t - 4$. Adding 4 to both sides of this equation yields $8 = t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

Q3

18

The correct answer is 18. Subtracting 45 from each side of the given equation yields $x - 9 = 18$. By the definition of absolute value, if $x - 9 = 18$, then $x - 9 = 18$ or $x - 9 = -18$. Adding 9 to each side of the equation $x - 9 = 18$ yields $x = 27$. Adding 9 to each side of the equation $x - 9 = -18$ yields $x = -9$. Therefore, the solutions to the given equation are 27 and -9, and it follows that the sum of the solutions to the given equation is $27 + -9$, or 18.

Q4

48

The correct answer is 48. It's given that in the xy-plane, when the graph of the function f, where $y = fx$, is shifted up 6 units, the resulting graph is defined by the function g. Therefore, function g can be defined by the equation $gx = fx + 6$. It's given that $fx = x - 1x + 3x - 2$. Substituting $x - 1x + 3x - 2$ for fx in the equation $gx = fx + 6$ yields $gx = x - 1x + 3x - 2 + 6$. For the point $4, b$, the value of x is 4. Substituting 4 for x in the equation $gx = x - 1x + 3x - 2 + 6$ yields $g4 = 4 - 14 + 34 - 2 + 6$, or $g4 = 48$. It follows that the graph of $y = gx$ crosses through the point $4, 48$. Therefore, the value of b is 48.

Q5

B

B. 10

Choice B is correct. Adding 9 to each side of the given equation yields $x^2 - 2x = 9$. To complete the square, adding 1 to each side of this equation yields $x^2 - 2x + 1 = 9 + 1$, or $x - 1^2 = 10$. Taking the square root of each side of this equation yields $x - 1 = \pm\sqrt{10}$. Adding 1 to each side of this equation yields $x = 1 \pm \sqrt{10}$. Since it's given that one of the solutions to the equation can be written as $1 + \sqrt{k}$, the value of k must be 10.

Alternate approach: It's given that $1 + \sqrt{k}$ is a solution to the given equation. It follows that $x = 1 + \sqrt{k}$. Substituting $1 + \sqrt{k}$ for x in the given equation yields $1 + \sqrt{k}^{\,2} - 21 + \sqrt{k} - 9 = 0$, or $1 + \sqrt{k}\ 1 + \sqrt{k} - 21 + \sqrt{k} - 9 = 0$. Expanding the products on the left-hand side of this equation yields $1 + 2\sqrt{k} + k - 2 - 2\sqrt{k} - 9 = 0$, or $k - 10 = 0$. Adding 10 to each side of this equation yields $k = 10$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

D

D. $w = 94^{n-1}$

Choice D is correct. Since w represents the nth term of the sequence and 9 is the first term of the sequence, the value of w is 9 when the value of n is 1. Since each term after the first is 4 times the preceding term, the value of w is 94 when the value of n is 2. Therefore, the value of w is 944, or 94^2, when the value of n is 3. More generally, the value of w is 94^{n-1} for a given value of n. Therefore, the equation $w = 94^{n-1}$ gives w in terms of n.

Choice A is incorrect. This equation describes a sequence for which the first term is 36, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is 4, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is 36, rather than 9.

Q7

C

C. $-\frac{319}{4}$

Choice C is correct. In the *xy*-plane, the graph of the line $y = c$ is a horizontal line that crosses the *y*-axis at $y = c$ and the graph of the quadratic equation $y = -x^2 + 9x - 100$ is a parabola. A parabola can intersect a horizontal line at exactly one point only at its vertex. Therefore, the value of c should be equal to the *y*-coordinate of the vertex of the graph of the given equation. For a quadratic equation in vertex form, $y = ax - h^2 + k$, the vertex of its graph in the *xy*-plane is h, k. The given quadratic equation, $y = -x^2 + 9x - 100$, can be rewritten as $y = -x^2 - 2\frac{9}{2}x + \frac{9}{2}^2 + \frac{9}{2}^2 - 100$, or $y = -x - \frac{9}{2}^2 + -\frac{319}{4}$. Thus, the value of c is equal to $-\frac{319}{4}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

D

D. $Vx = 9xx - 7$

Choice D is correct. The volume of a right rectangular prism can be represented by a function V that gives the volume of the prism, in cubic inches, in terms of the length of the prism's base. The volume of a right rectangular prism is equal to the area of its base times its height. It's given that the length of the prism's base is x inches, which is 7 inches more than the width of the prism's base. This means that the width of the prism's base is $x - 7$ inches. It follows that the area of the prism's base, in square inches, is $xx - 7$ and the volume, in cubic inches, of the prism is $xx - 79$. Thus, the function V that gives the volume of this right rectangular prism, in cubic inches, in terms of the length of the prism's base, x, is $Vx = 9xx - 7$.

Choice A is incorrect. This function would give the volume of the prism if the height were 9 inches more than the length of its base and the width of the base were 7 inches more than its length.

Choice B is incorrect. This function would give the volume of the prism if the height were 9 inches more than the length of its base.

Choice C is incorrect. This function would give the volume of the prism if the width of the base were 7 inches more than its length, rather than the length of the base being 7 inches more than its width.

Q9

C

C. $\frac{1}{2}$

Choice C is correct. Each fraction in the given equation can be expressed with the common denominator x^2-9. Multiplying $\frac{2x}{x+3}$ by $\frac{x-3}{x-3}$ yields $\frac{2x^2-6x}{x^2-9}$, and multiplying $\frac{1}{x-3}$ by $\frac{x+3}{x+3}$ yields $\frac{x+3}{x^2-9}$. Therefore, the given equation can be written as $\frac{4x^2}{x^2-9}-\frac{2x^2-6x}{x^2-9}=\frac{x+3}{x^2-9}$. Multiplying each fraction by the denominator results in the equation $4x^2-(2x^2-6x)=x+3$, or $2x^2+6x=x+3$. This equation can be solved by setting a quadratic expression equal to 0, then solving for x. Subtracting $x+3$ from both sides of this equation yields $2x^2+5x-3=0$. The expression $2x^2+5x-3$ can be factored, resulting in the equation $(2x-1)(x+3)=0$. By the zero product property, $2x-1=0$ or $x+3=0$. To solve for x in $2x-1=0$, 1 can be added to both sides of the equation, resulting in $2x=1$. Dividing both sides of this equation by 2 results in $x=\frac{1}{2}$. Solving for x in $x+3=0$ yields $x=-3$. However, this value of x would result in the second fraction of the original equation having a denominator of 0. Therefore, $x=-3$ is an extraneous solution. Thus, the only value of x that satisfies the given equation is $x=\frac{1}{2}$.

Choice A is incorrect and may result from solving $x+3=0$ but not realizing that this solution is extraneous because it would result in a denominator of 0 in the second fraction. Choice B is incorrect and may result from a sign error when solving $2x-1=0$ for x. Choice D is incorrect and may result from a calculation error.

Q10 B

B. $-1 < r < 0$

Choice B is correct. The term $(1 + r)$ represents a percent change. Since the population is decreasing, the percent change must be between 0% and 100%. When the percent change is expressed as a decimal rather than as a percent, the percentage change must be between 0 and 1. Because $(1 + r)$ represents percent change, this can be expressed as $0 < 1 + r < 1$. Subtracting 1 from all three terms of this compound inequality results in $-1 < r < 0$.

Choice A is incorrect. If $r < -1$, then after 1 year, the population P would be a negative value, which is not possible. Choices C and D are incorrect. For any value of $r > 0$, $1 + r > 1$, and the exponential function models growth for positive values of the exponent. This contradicts the given information that the population is decreasing.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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